

Problem

Two coils connected in series-aiding fashion have a total inductance of 500 mH. When connected in a series-opposing configuration, the coils have a total inductance of 300 mH. If the inductance of one coil (L_1) is three times the other, find L_1 , L_2 , and M . What is the coupling coefficient?

Step-by-step solution

Step 1 of 5

Write the expression for equivalent inductance (L_{eq1}) when two coils are in series-aided.

$$L_{eq1} = L_1 + L_2 + 2M \quad \dots\dots (1)$$

Here, L_1 is the self-inductance of inductor 1, L_2 is the self-inductance of inductor 2, and M is the mutual-inductance.

Substitute 500 mH for L_{eq1} in equation (1).

$$500 \text{ mH} = L_1 + L_2 + 2M \quad \dots\dots (2)$$

Write the expression for equivalent inductance (L_{eq2}) when two coils are in series-opposite.

$$L_{eq2} = L_1 + L_2 - 2M$$

Substitute 300 mH for L_{eq2} .

$$300 \text{ mH} = L_1 + L_2 - 2M \quad \dots\dots (3)$$

Step 2 of 5

Add equation (1) and equation (2).

$$L_1 + L_2 - 2M + L_1 + L_2 + 2M = 500 \text{ mH} + 300 \text{ mH}$$

$$2L_1 + 2L_2 = 800 \text{ mH}$$

Since $L_1 = 3L_2$, therefore,

$$2(3L_2) + 2L_2 = 800 \text{ mH}$$

$$8L_2 = 800 \text{ mH}$$

$$L_2 = \frac{800 \text{ mH}}{8}$$

$$= 100 \text{ mH}$$

Hence, the value of inductance L_2 is 100 mH.

Step 3 of 5

Since $L_1 = 3L_2$, substitute 100 mH for L_2 .

$$L_1 = 3(100 \text{ mH})$$

$$= 300 \text{ mH}$$

Hence, the value of the inductance L_1 is 300 mH.

Step 4 of 5

Determine the value of the mutual inductance.

Substitute 100 mH for L_2 and 300 mH for L_1 in the equation (2).

$$500 \text{ mH} = 300 \text{ mH} + 100 \text{ mH} + 2M$$

$$2M = 500 \text{ mH} - 400 \text{ mH}$$

$$2M = 100 \text{ mH}$$

$$M = \frac{100 \text{ mH}}{2}$$

$$= 50 \text{ mH}$$

Hence, the value of the mutual inductance M is 50 mH.

Step 5 of 5

The coefficient of coupling is as follows:

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

Substitute 100 mH for L_2 , 300 mH for L_1 , and 50 mH for M .

$$k = \frac{50 \times 10^{-3}}{\sqrt{(100 \times 10^{-3})(300 \times 10^{-3})}}$$

$$= \frac{50 \times 10^{-3}}{173.20 \times 10^{-3}}$$

$$\cong 0.2887$$

Hence, the value of coefficient of coupling k is 0.2887.